Technical report: Bon·Appétit

Swiss supermarket receipts are unreadable to software: lines are truncated, multilingual and coded, and the Migros export has no barcodes. Bon·Appétit uses Apertus v1.5 8B, running on the household's own laptop, to read each line and link it to the federal Swiss Food Composition Database, with a symbolic food layer that checks every pick and code that does every number. On 160 real Swiss products it has 41% less energy error than text matching and, on a held-out half, 26% less than Apertus guessing alone (paired 95% CI excludes zero), and every number it reports traces to a federal food record.

- **Track:** Track 2B: Bon·Appétit
- **Event:** Online
- **Team:** Dhiraj Kavuri (solo)
- **Demo:** video linked in the submission form · live page https://kinnuworks.github.io/bon-appetit/
- **Code:** https://github.com/kinnuworks/bon-appetit

1. Summary

Households, dietitians and prevention programmes already hold a detailed record of what people eat: their supermarket receipts. They cannot use it, because a line such as `QP GRUY.RAPE 200G` means nothing to a lookup table, and the purchase history is too sensitive to send to a foreign cloud model. Bon·Appétit reads receipts with Apertus on the household's own computer, links each line to the FSVO database, abstains when unsure, and turns a week of shopping into grams per adult per day for each group of the Swiss food pyramid. On 160 real Swiss products with printed nutrition labels, Bon·Appétit's energy estimate is off by 102 kcal per 100 g on average, 41% less than fuzzy text matching (171) and less than Apertus guessing the numbers itself (122), which gives no source. On the held-out half, where nothing was tuned, the gap to Apertus guessing is 26%. Only 34% of products are off by more than 100 kcal, against 54% for text matching. Both Apertus steps matter: the pick alone cuts search error by 19% (153 to 123), and the cross-check against the model's own estimate takes it to 102, while every reported number still comes from a federal food record.

2. Architecture

```
receipt (CSV, PDF, text) → parser (rules) → Apertus: read line → search (lexical + Swiss embeddings)
   → Apertus: pick or abstain → Apertus: own estimate, cross-check among top candidates → food layer (unit, energy, pack, Swiss words) → ledger (code) → Apertus: one nudge (slots)
```

Component	What it does	Where it runs
Parser	Prices, quantities, pack sizes, multipacks, non-food hints	Python, rules
Read	One Apertus call: English description, language, food or not, generic German food name	Apertus v1.5 8B
Search	Reciprocal-rank fusion of token-set matching and an Apertus-derived Swiss embedding model over 1,145 FSVO foods	CPU or Apple GPU
Pick	One Apertus call: list number of the best candidate, or 0	Apertus v1.5 8B
Cross-check	One Apertus call: the model's own per-100 g estimate; code keeps the top-8 candidate whose FSVO record is closest, with a head start for the model's pick. The estimate is never shown	Apertus v1.5 8B + code
Food layer	Vetoes physically impossible links; Swiss food-word glossary; SGE food groups	Python, rules
Ledger	Grams per adult per day, sugar, salt, saturated fat, against SGE-derived references	Python

Component	What it does	Where it runs
Nudge	One sentence in DE/FR/IT/EN; numbers enter only as placeholders filled by code	Apertus v1.5 8B

Target architecture

b) Air-gapped, also **a) on-premise**. `make airgap` runs the app and a llama.cpp server on a Docker network marked `internal`, and the app runs an egress test before it starts (it tries four outside hosts and refuses to start if any answers). Build time needs the internet once: Docker images, the GGUF model file, the embedding model, the Open Food Facts dump for evaluation. Runtime needs none. A Swiss cloud endpoint works through `LLM_BASE_URL` (option c).

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-8B`, as the community text-only Q4_K_M GGUF `Colby/apertus-v1.5-8b-text-Q4_K_M-GGUF`. The text-only conversion follows the FHNW conversion that organisers accepted on Discord for Track 2B.
- **How it is used:** inference, three calls per receipt line (read, pick, estimate for the cross-check) plus one per nudge; thinking off; temperature 0; seed 0; JSON output.
- **Where it runs:** local weights through Ollama 0.40 (Apple M5, 16 GB) for development and evaluation; llama.cpp server in Docker for `make local` and `make airgap`.
- **Other open models:** `andreasmartin/apertus-v1.1-swiss-embed-0.4b-bidir` (Apache-2.0), an embedding model derived from Apertus, for candidate search. No closed model is used at runtime or in evaluation.

Two choices came from measurement. A full JSON-schema grammar on llama.cpp made each call about four times slower and made the 8B model answer "no match" for almost every line, including easy ones such as bananas; plain JSON mode with the schema enforced in code fixed both. The model also reports "not sure" for nearly every pick, so that flag is recorded but does not decide anything; abstention comes from choosing 0, and review from the food layer.

4. Data

Source	Use	Licence
FSVO Swiss Food Composition Database V6.5 (1,109 generic foods, 36 brand products)	The food catalogue and every nutrient number	Free reuse with attribution
Open Food Facts product database (parquet dump, October 2026)	Evaluation: Swiss products with printed nutrition values	ODbL-1.0; barcodes and label values for the 160 evaluation products are kept under ODbL, the dump is not redistributed
SGE Swiss food pyramid portions	Daily reference amounts per food group	Cited guidance

No personal data is used. No real receipts were available under a licence, so receipt lines in the evaluation are produced from real product names by documented rules (`eval/build_set.py`): shop-brand shorthand (M-Budget becomes `M-BUDG`, Qualité & Prix becomes `QP`), accent folding, upper case, long words cut to seven letters plus a dot, 24 characters, pack size appended.

5. Evaluation

Task. For each product, hide the printed label, give the system only the receipt-style line, and compare its per-100 g estimate with the printed values. A method that links to an FSVO food is scored with that food's values.

Set. 160 products were sampled with seed 42 from the 74,192 Open Food Facts entries sold in Switzerland with complete nutrition tables (80 German, 50 French, 30 Italian names). The model runs at about 8 tokens per second on the laptop, so the run was stopped at the deadline after 160 products. Languages were interleaved in the run

order, so the scored products keep the same language mix. The glossary and shortening rules were written before this set was built.

Methods. 1. Fuzzy text match: the best token-set match among FSVO names, the approach of earlier receipt tools. 2. Swiss embeddings only: nearest FSVO food by embedding. 3. Search only (ablation): the top candidate of the hybrid's own search, with no Apertus pick. The difference to the hybrid is what the pick step adds. 4. Apertus alone: the model estimates the nutrients directly from the line, with no database and no source. 5. Hybrid, Apertus pick only. 6. Bon·Appétit: pick plus the cross-check against Apertus's own estimate. The two cross-check settings were chosen on half of the products and are reported separately on the other half. 7. Label-informed ceiling: the FSVO food closest to the printed label, chosen by looking at the label. No real system can do this; it shows how close any link to generic FSVO foods could get.

The hybrid abstains on a few lines; the second table compares every method on exactly the products the hybrid answered.

Method	kcal error per 100 g (95% CI)	Off by >100 kcal	Sugar error, g	Salt error, g	Food group right	Abstains
Fuzzy text match	171 (151 to 195)	54%	13.6	0.85	27%	0%
Swiss embeddings only	194 (171 to 219)	62%	17.9	0.85	27%	0%
Search only, no Apertus pick (ablation)	153 (129 to 177)	49%	12.3	0.92	38%	5%
Apertus alone, guessing (no source)	122 (104 to 144)	44%	10.3	0.69	n/a	6%
Hybrid, Apertus pick only	123 (102 to 145)	39%	10.7	0.69	49%	8%
Bon·Appétit: pick plus Apertus cross-check	102 (84 to 118)	34%	9.0	0.68	50%	8%
Label-informed ceiling (best FSVO food)	12 (10 to 15)	0%	1.2	0.66	61%	0%

Held-out check. The cross-check has two settings (how many candidates it looks at, and how strongly it favours the model's own pick). They were chosen on the odd-numbered products only; this table scores every method on the even-numbered products, which played no part in that choice.

Method, held-out half only	n	kcal error (95% CI)
Fuzzy text match	80	162 (131 to 194)
Search only, no Apertus pick (ablation)	76	144 (115 to 176)
Apertus alone, guessing (no source)	74	128 (100 to 157)
Hybrid, Apertus pick only	73	120 (92 to 152)
Bon·Appétit: pick plus Apertus cross-check	73	94 (71 to 124)
Label-informed ceiling (best FSVO food)	80	11 (9 to 14)

Method, on the products the hybrid answered	n	kcal error (95% CI)	Off by >100 kcal
Fuzzy text match	148	172 (149 to 197)	53%
Swiss embeddings only	148	196 (170 to 221)	62%
Search only, no Apertus pick (ablation)	148	153 (129 to 176)	50%
Apertus alone, guessing (no source)	140	116 (98 to 137)	41%
Hybrid, Apertus pick only	148	123 (102 to 145)	39%
Bon·Appétit: pick plus Apertus cross-check	148	102 (84 to 118)	34%
Label-informed ceiling (best FSVO food)	148	12 (10 to 15)	0%

Running the same energy check over all 86,517 Swiss products on Open Food Facts flagged 3,035 (4%) whose printed calories contradict their own macronutrients; 433 look like kilojoules typed as kilocalories. The list is in `eval/off_energy_flags.csv`. The same rule flags none of the 1,046 non-alcoholic foods in the federal FSVO database, so it is not firing at random: the inconsistencies are in crowd-sourced entries, where a volunteer typed the label.

Paired comparison on the same products (Bon·Appétit minus the other method; negative means Bon·Appétit is closer to the label). Against text matching: -71 kcal (95% CI -98 to -44, n=148). Against the Apertus pick without the cross-check: -22 kcal (95% CI -40 to -4, n=148), so the cross-check helps. Against Apertus guessing alone, on the held-out half: -26 kcal (95% CI -54 to -1, n=68); on the full set the difference is -13 kcal (95% CI -31 to +4, n=140) and not statistically clear. The firm claims are therefore the gains over text matching and over the plain pick, plus the held-out gain over guessing; the full-set gap over guessing is a trend.

Household view. Over 200 simulated weekly baskets of 20 products, pyramid shares are off by 7.3 percentage points per group (text matching 8.5).

By language (Bon·Appétit = pick plus cross-check). German and French gain clearly. Italian does not: with 30 products it is no better than Apertus guessing, and only a quarter land in the right food group. The Italian sample is small, and Italian shop shorthand is the next thing to work on.

Language	n	Fuzzy kcal error	Embeddings kcal error	Apertus alone kcal error	Bon·Appétit kcal error (pick + cross-check)	Bon·Appétit food group right
DE	80	164	216	123	95	59%
FR	50	183	179	112	92	52%
IT	30	170	161	137	140	25%

Reproduce. `make eval` re-scores the stored predictions without a model and checks the SHA-256 of the results against `eval/results.sha256`. `python -m eval.run` regenerates the predictions with any Apertus endpoint; every model answer is also stored in `data/llm_cache.jsonl`.

6. Value and cost

Measured on an Apple M5 laptop with 16 GB, Apertus v1.5 8B Q4_K_M through Ollama, four requests in parallel: 47.1 seconds per receipt line for all three Apertus calls together, including queueing. A weekly receipt of 20 food lines therefore takes about 15.7 minutes on one laptop, with no subscription and no per-request fee. Electricity for 1,000 such receipts is about CHF 2.36 (laptop at 30 W, 20 food lines per receipt, CHF 0.30 per kWh, measured seconds per line with 4 parallel requests). The laptop run is a demonstration of the sovereign setup, not of speed. Throughput on a server GPU or a Swiss cloud endpoint was not measured here; that is the path for a retailer-scale service and the first thing a pilot would measure.

Per receipt line the run used about 581 input and 107 output tokens over three Apertus calls. At the measured laptop rate that is about 76 lines per hour per laptop with four requests in parallel. Server throughput was not measured.

The users are households first, then dietitians, home-care services and prevention programmes working with a client's consent, and retailers who want a nutrition view inside a loyalty app without sending purchase data to a foreign model. A first pilot needs 20 consenting households, four weeks of receipts, a dietitian's review of the reference amounts, and one outcome read from the receipts themselves: the change in the vegetables-and-fruit share. The FoodCoach trial (NCT05597969) is the reference design. No effect is claimed yet.

7. Limitations

- Purchases are not intake: waste, guests and meals out are invisible. The output is a household trend by food group, not a per-person diet record.
- The receipt lines are synthetic, built from real names by stated rules. Real tills cut and code differently; the review queue and a household alias list are meant to absorb this.
- FSVO has generic foods, so a branded product is scored against a generic value. The oracle row shows how much error that alone causes.

- One run, one seed, one quantisation. 70B was not evaluated because hackathon inference keys were exhausted.
- Photo receipts are not supported yet; Apertus image input was not used, because the community GGUF builds are text-only.
- Reference amounts per food group are derived by hand from SGE portions and are illustrative until a dietitian reviews them. Lines without a pack size get a default amount per group, counted as estimated.
- The cross-check row is computed by `eval/score.py` from the stored model answers: the "Apertus alone" estimate in the evaluation uses the same prompt as the live cross-check call, so the scored decision is the one the app makes. Two other linker changes came after the run (non-food words always win; the energy check on the chosen FSVO food); neither applies to the evaluation products, which are all foods with consistent FSVO records.
- The cross-check's two settings were chosen by trying 12 combinations on the odd-numbered products. The held-out table scores the even-numbered ones only; the full-set row includes the half the settings were chosen on.

8. Lessons on building with Apertus 1.5

- The 8B model reads shop shorthand well (`M-BUDG VOLLMILCH UHT 1L` becomes "M-Budget whole milk, UHT, 1 litre") but misses Swiss food words: it read `RUEBLI` (carrot) as kale and as flatbread. A twenty-line glossary in the food layer fixed it; the words belong in localisation data for the next Apertus release.
- Asked for Swiss spelling, the 8B model still wrote "ß" in German suggestions ("Süßigkeiten"). Code now converts it, as Swiss orthography always uses "ss".
- Without the target language's food-group names in the prompt, it mixed English group names into German and French text, and once called a household "doing great with sweets" at five times the guide. Giving it the direction ("too little", "too much") and the names in the target language fixed both.
- Constrained decoding with a full schema hurt accuracy as well as speed on llama.cpp.
- The model is cautious when it chooses from a list and almost never says it is sure. Selection by list number plus external checks worked better than asking for confidence.
- Apertus alone produced plausible numbers (122 kcal mean error) with no source. Used as a cross-check on its own pick instead of as the answer, the same estimate made the linked results better than either alone (102).

9. Next steps

Real receipts donated with consent (a consent template is in `docs/`); Coop Supercard export; a household alias memory shared across weeks; the 70B model for the read step on a Swiss cloud; reporting nutrient ranges instead of point values; a contribution back to Open Food Facts listing Swiss entries whose printed values fail the Atwater check.

License

Creative Commons Attribution 4.0 (CC-BY-4.0). All Hack Apertus projects are open-sourced. Code: Apache-2.0.

References

- FSVO, Swiss Food Composition Database V6.5, naehrwertdaten.ch, 2023.
- Swiss Society for Nutrition (SGE), Swiss food pyramid, sge-ssn.ch.
- Open Food Facts, product database, ODbL.
- Apertus v1.5 model card, huggingface.co/swiss-ai/Apertus-v1.5-8B.
- A. Martin, apertus-v1.1-swiss-embed-0.4b-bidir, Hugging Face.
- FoodCoach trial, ClinicalTrials.gov NCT05597969; Receipt2Nutrition, ETH Zurich.
- TooSweet, Open Food Hackdays 2018, hack.opendata.ch/project/153.